

Introduction to Abstract Algebra II: Optional Homework

Due on March 17, 2026 at 23:59

Victor Ostrik 10:00

Hashem A. Damrah

UO ID: 952102243

Exercise 22.46. An element a of a ring R is idempotent if $a^2 = a$.

- (i) Show that the set of all idempotent elements of a commutative ring is closed under multiplication.
- (ii) Find all idempotents in the ring $\mathbb{Z}_6 \times \mathbb{Z}_{12}$.

Solution to (i). Let R be a commutative ring, and let $a, b \in R$ be idempotent elements. Then

$$a^2 = a \quad \text{and} \quad b^2 = b.$$

We want to show that ab is also idempotent, i.e., that $(ab)^2 = ab$. Using associativity and commutativity of R , we compute:

$$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = (aa)(bb) = a^2b^2.$$

Since $a^2 = a$ and $b^2 = b$, this becomes $(ab)^2 = ab$. Thus, ab is idempotent. Therefore, the set of all idempotent elements of a commutative ring is closed under multiplication. $\square$

Solution to (ii). In the direct product ring $\mathbb{Z}_6 \times \mathbb{Z}_{12}$, an element (a, b) is idempotent if and only if

$$(a, b)^2 = (a, b),$$

which is equivalent to

$$(a^2, b^2) = (a, b)$$

Thus, $a^2 \equiv a \pmod{6}$ and $b^2 \equiv b \pmod{12}$.

We solve $a^2 \equiv a \pmod{6}$, i.e., $a(a-1) \equiv 0 \pmod{6}$. Checking $a = 0, 1, 2, 3, 4, 5$:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 3^2 = 9 \equiv 3, \quad 4^2 = 16 \equiv 4 \pmod{6}.$$

Thus, the idempotents in $\mathbb{Z}_6$ are $\{0, 1, 3, 4\}$.

Next, we solve $b^2 \equiv b \pmod{12}$, i.e., $b(b-1) \equiv 0 \pmod{12}$. Checking $b = 0, 1, \dots, 11$:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 4^2 = 16 \equiv 4, \quad 9^2 = 81 \equiv 9 \pmod{12}.$$

Thus, the idempotents in $\mathbb{Z}_{12}$ are $\{0, 1, 4, 9\}$.

An element (a, b) is idempotent if and only if a is idempotent in $\mathbb{Z}_6$ and b is idempotent in $\mathbb{Z}_{12}$. Hence, all idempotents are:

$$\{0, 1, 3, 4\} \times \{0, 1, 4, 9\}. \quad \square$$

Exercise 22.48. An element a of a ring R is nilpotent if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that if a and b are nilpotent elements of a commutative ring, then $a + b$ is also nilpotent.

Solution. Let R be a commutative ring, and suppose that $a, b \in R$ are nilpotent. Then there exist positive integers $m, n \in \mathbb{Z}^+$ such that

$$a^m = 0 \quad \text{and} \quad b^n = 0.$$

We will show that $a + b$ is nilpotent.

Consider $(a + b)^{m+n}$. By the binomial theorem (which applies since R is commutative), we have

$$(a + b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} a^k b^{m+n-k}.$$

Now consider any term $a^k b^{m+n-k}$ in this sum. Either $k \geq m$ or $m+n-k \geq n$. If $k \geq m$, then $a^k = 0$ since $a^m = 0$. If $k < m$, then $m+n-k > n-1$, so $m+n-k \geq n$, and hence $b^{m+n-k} = 0$ since $b^n = 0$. Thus, every term in the sum is zero, so $(a + b)^{m+n} = 0$. Therefore, $a + b$ is nilpotent. $\square$

Exercise 22.49. Show that a ring R has no nonzero nilpotent element if and only if 0 is the only solution of $x^2 = 0$ in R .

Solution. Suppose that R has no nonzero nilpotent elements. Let $x \in R$ satisfy $x^2 = 0$. Then x is nilpotent (with $n = 2$), so by assumption we must have $x = 0$. Hence, 0 is the only solution of $x^2 = 0$ in R .

Conversely, suppose that 0 is the only solution of $x^2 = 0$ in R . Let $a \in R$ be nilpotent, so that $a^n = 0$ for some $n \in \mathbb{Z}^+$. We will show that $a = 0$. Choose n minimal such that $a^n = 0$. If $n = 1$, then $a = 0$, and we are done. Suppose $n > 1$. Then $a^{n-1} \neq 0$ by minimality of n , and

$$(a^{n-1})^2 = a^{2n-2} = 0,$$

since $2n - 2 \geq n$ and $a^n = 0$ implies all higher powers are also 0. Thus, a^{n-1} is a nonzero element satisfying $(a^{n-1})^2 = 0$, contradicting the assumption that 0 is the only solution of $x^2 = 0$. Therefore, no nonzero nilpotent elements exist in R . $\square$

Exercise 22.50. Show that a subset S of a ring R gives a subring of R if and only if the following hold:

$$\begin{aligned} 0 &\in S; \\ (a - b) &\in S \text{ for all } a, b \in S; \\ ab &\in S \text{ for all } a, b \in S. \end{aligned}$$

Solution. We prove the statement by showing both directions.

Suppose that S is a subring of R . Then, by definition of a subring:

- $0 \in S$.
- If $a, b \in S$, then $a - b \in S$ (since S is closed under addition and additive inverses).
- If $a, b \in S$, then $ab \in S$ (since S is closed under multiplication).

Thus, all three conditions hold.

Conversely, suppose that $S \subseteq R$ satisfies:

- $0 \in S$,
- $a - b \in S$ for all $a, b \in S$,
- $ab \in S$ for all $a, b \in S$.

We show that S is a subring of R . First, we show closure under addition. Let $a, b \in S$. Since $b \in S$, we have $0 - b = -b \in S$. Then

$$a + b = a - (-b),$$

and since $a, -b \in S$, it follows that $a + b \in S$. Next, S is closed under additive inverses since for any $a \in S$,

$$-a = 0 - a \in S.$$

Closure under multiplication is given by assumption. Thus, S is closed under addition, additive inverses, and multiplication, and contains 0. Therefore, S is a subring of R . $\square$

Exercise 22.57. A ring R is a Boolean ring if $a^2 = a$ for all $a \in R$, so that every element is idempotent. Show that every Boolean ring is commutative.

Solution. Let R be a Boolean ring, so that $a^2 = a$ for all $a \in R$. We will show that R is commutative. First, observe that for any $a \in R$,

$$(a + a)^2 = a + a.$$

Expanding the left-hand side, we get

$$(a + a)^2 = a^2 + aa + aa + a^2 = a + a + a + a = 4a.$$

Thus, $4a = 2a$. So $a + a = 0$ for all $a \in R$. Now let $a, b \in R$. Then

$$(a + b)^2 = a + b.$$

Expanding, we obtain $(a + b)^2 = a + ab + ba + b$. Thus, $a + ab + ba + b = a + b$. Canceling $a + b$ from both sides, we get $ab + ba = 0$. But since $a + a = 0$ for all a , we have $x = -x$ for all $x \in R$. Hence,

$$ab = -ab = ba.$$

Therefore, $ab = ba$ for all $a, b \in R$, so R is commutative. $\square$

Exercise 24.29. Use Fermat's theorem to show that for any positive integer n , the integer $n^{37} - n$ is divisible by 383838. [Hint: $383838 = (37)(19)(13)(7)(3)(2)$.]

Solution. We will show that $n^{37} - n$ is divisible by each of the primes 2, 3, 7, 13, 19, 37, and since these are distinct, their product

$$383838 = 2 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37,$$

will also divide $n^{37} - n$.

Let p be one of these primes. We consider two cases. If $p \mid n$, then $n \equiv 0 \pmod{p}$, so

$$n^{37} - n \equiv 0 - 0 \equiv 0 \pmod{p}.$$

If $\gcd(n, p) = 1$, then by Fermat's Little Theorem,

$$n^{p-1} \equiv 1 \pmod{p}.$$

Since $p - 1 \mid 36$ for each $p \in \{2, 3, 7, 13, 19, 37\}$, we can write $36 = k(p - 1)$ for some integer k . Then

$$n^{36} = (n^{p-1})^k \equiv 1^k \equiv 1 \pmod{p}.$$

Multiplying both sides by n , we obtain

$$n^{37} \equiv n \pmod{p},$$

so

$$n^{37} - n \equiv 0 \pmod{p}.$$

Thus, for every prime $p \in \{2, 3, 7, 13, 19, 37\}$, we have

$$p \mid (n^{37} - n).$$

Since these primes are distinct, their product divides $n^{37} - n$. Therefore,

$$383838 \mid (n^{37} - n). \quad \square$$

Exercise 24.30. Referring to Exercise 29, find a number larger than 383838 that divides $n^{37} - n$ for all positive integers n .

Solution. From Exercise 29, we know that $383838 = 2 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37$, divides $n^{37} - n$ for all integers n . To find a larger integer with the same property, we check whether higher powers of these primes also divide $n^{37} - n$ for all n . Note that if n is even, then n^{37} is even, so $n^{37} - n$ is even. If n is odd, then n^{37} is odd, so $n^{37} - n$ is even. Thus,

$$2 \mid (n^{37} - n) \quad \text{for all } n.$$

But, we can say an even stronger statement. If n is even, both n^{37} and n are divisible by 4 or congruent to 0 or 2 module 4. You can check this directly that $n^{37} \equiv n \pmod{4}$. If n is odd, then $n \equiv 1$ or $3 \pmod{4}$, and in either case, $n^{37} \equiv n \pmod{4}$. Thus,

$$4 \mid (n^{37} - n) \quad \text{for all } n.$$

However, no higher power of 2 works in general. For example, taking $n = 2$, we have

$$2^{37} - 2 \equiv 6 \pmod{8},$$

so $8 \nmid (n^{37} - n)$ in general.

For the odd primes 3, 7, 13, 19, 37, one can check that higher powers (such as p^2) do not divide $n^{37} - n$ for all n . Thus, we cannot increase their exponents. Therefore, the largest integer of this form is obtained by replacing 2 with 4:

$$4 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37 = 767676.$$

Hence, a number larger than 383838 that divides $n^{37} - n$ for all positive integers n is 767676. $\square$

Exercise 27.32. Let $\varphi : R_1 \rightarrow R_2$ be a ring homomorphism. Show that there is a unique ring homomorphism $\psi : R_1[x] \rightarrow R_2[x]$ such that $\psi(a) = \varphi(a)$ for any $a \in R_1$ and $\psi(x) = x$.

Solution. Let $\varphi : R_1 \rightarrow R_2$ be a ring homomorphism. We will construct a ring homomorphism $\psi : R_1[x] \rightarrow R_2[x]$, such that $\psi(a) = \varphi(a)$ for all $a \in R_1$ and $\psi(x) = x$, and then show it is unique.

Firstly, we prove its existence. Every polynomial $f(x) \in R_1[x]$ can be written uniquely as

$$f(x) = a_0 + a_1x + \cdots + a_nx^n,$$

with $a_i \in R_1$. Define $\psi : R_1[x] \rightarrow R_2[x]$ by

$$\psi(f(x)) = \varphi(a_0) + \varphi(a_1)x + \cdots + \varphi(a_n)x^n.$$

That is, ψ applies φ to each coefficient. Clearly, for $a \in R_1$ (viewed as a constant polynomial), $\psi(a) = \varphi(a)$, and $\psi(x) = x$.

We now check that ψ is a ring homomorphism. For $f(x) = \sum a_i x^i$ and $g(x) = \sum b_i x^i$,

$$\psi(f + g) = \sum \varphi(a_i + b_i)x^i = \sum (\varphi(a_i) + \varphi(b_i))x^i = \psi(f) + \psi(g).$$

Using the Cauchy product,

$$fg = \sum_k \left(\sum_{i+j=k} a_i b_j \right) x^k,$$

so

$$\psi(fg) = \sum_k \varphi \left(\sum_{i+j=k} a_i b_j \right) x^k = \sum_k \left(\sum_{i+j=k} \varphi(a_i) \varphi(b_j) \right) x^k = \psi(f) \psi(g).$$

Thus, ψ is a ring homomorphism.

Next, we prove uniqueness. Suppose $\psi' : R_1[x] \rightarrow R_2[x]$ is another ring homomorphism such that $\psi'(a) = \varphi(a)$ for all $a \in R_1$ and $\psi'(x) = x$. For any polynomial $f(x) = \sum a_i x^i$, we have

$$\psi'(f(x)) = \sum \psi'(a_i) \psi'(x)^i = \sum \varphi(a_i) x^i = \psi(f(x)).$$

Thus, $\psi' = \psi$, proving uniqueness.

Therefore, there exists a unique ring homomorphism $\psi : R_1[x] \rightarrow R_2[x]$ such that $\psi(a) = \varphi(a)$ for all $a \in R_1$ and $\psi(x) = x$. $\square$

Exercise 28.38. Find all irreducible polynomials of degree 3 in $\mathbb{Z}_2[x]$.

Solution. We work in $\mathbb{Z}_2[x]$, where $\mathbb{Z}_2 = \{0, 1\}$. A polynomial of degree 3 has the form

$$f(x) = x^3 + ax^2 + bx + c, \quad a, b, c \in \mathbb{Z}_2.$$

Since the leading coefficient must be 1, there are $2^3 = 8$ such polynomials.

A polynomial of degree 3 over a field is reducible if and only if it has a root in that field. Thus, $f(x)$ is reducible over $\mathbb{Z}_2$ if and only if $f(0) = 0$ or $f(1) = 0$. We list all degree 3 polynomials and test for roots:

$f(x)$	$f(0)$	$f(1)$
x^3	0	1
$x^3 + 1$	1	0
$x^3 + x$	0	0
$x^3 + x + 1$	1	1
$x^3 + x^2$	0	0
$x^3 + x^2 + 1$	1	1
$x^3 + x^2 + x$	0	1
$x^3 + x^2 + x + 1$	1	0

A polynomial is irreducible if it has no roots in $\mathbb{Z}_2$, so we look for those with $f(0) \neq 0$ and $f(1) \neq 0$. From the table, these are:

$$x^3 + x + 1 \quad \text{and} \quad x^3 + x^2 + 1.$$

$\square$

Exercise 28.39. Find all irreducible polynomials of degree 3 in $\mathbb{Z}_3[x]$.

Solution. We work in $\mathbb{Z}_3[x]$, where $\mathbb{Z}_3 = \{0, 1, 2\}$. A polynomial of degree 3 has the form

$$f(x) = x^3 + ax^2 + bx + c, \quad a, b, c \in \mathbb{Z}_3,$$

and there are $3^3 = 27$ such polynomials. Over a field, a polynomial of degree 3 is reducible if and only if it has a root in that field. Thus, $f(x)$ is irreducible over $\mathbb{Z}_3$ if and only if

$$f(0) \neq 0, \quad f(1) \neq 0, \quad f(2) \neq 0.$$

We first eliminate all polynomials with a root at 0. We have $f(0) = c$, so $c \neq 0$. Thus $c \in \{1, 2\}$. This leaves $3 \cdot 3 \cdot 2 = 18$ polynomials. Then, we eliminate polynomials with a root at 1. We compute

$$f(1) = 1 + a + b + c \not\equiv 0 \pmod{3}.$$

Next, a root at 2. Since $2^2 = 1$ and $2^3 = 2$ in $\mathbb{Z}_3$, we have

$$f(2) = 2 + a(1) + 2b + c = 2 + a + 2b + c \not\equiv 0 \pmod{3}.$$

We now check all $a, b \in \mathbb{Z}_3$ and $c \in \{1, 2\}$ satisfying both conditions. The resulting irreducible polynomials are:

$$x^3 + 2x + 1,$$

$$x^3 + x^2 + 2,$$

$$x^3 + x^2 + x + 2,$$

$$x^3 + x^2 + 2x + 1,$$

$$x^3 + 2x^2 + 1,$$

$$x^3 + 2x^2 + x + 1,$$

$$x^3 + 2x^2 + 2x + 2,$$

$$x^3 + 2x^2 + x + 2.$$

Thus, the irreducible polynomials of degree 3 in $\mathbb{Z}_3[x]$ are exactly the eight polynomials listed above. □